

Toward Quasi-Polynomial Unknot Recognition

Component Disc Certificates and Quadrilateral Core Reduction

Research continuation for Vladimir Reshetnikov’s ProveIt project

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Abstract

The maintained normal-surface backend computes aggregate component counts but does not decide whether a disconnected supplied surface contains a compressing disc. We implement and certify this missing operation without expanding binary normal coordinates. The main specialization uses only three additive orbit weights: Euler characteristic and two boundary-homology tests. A second reduction removes vertex-link components and divides by the content of the quadrilateral coordinates. We prove that the exact number of essential disc components scales linearly under this reduction, although the total number of surface components need not. The resulting orbit query depends on the bit size of the primitive quadrilateral data, independently of large additive triangle offsets. Independent arithmetic replay, explicit finite oracles, external Regina comparisons, and paired measurements accompany the implementation. A further proved two-weight variant uses certified ambient homology preprocessing; its implementation and timing are future work. The work is an algorithmic continuation of classical weighted Agol–Hass–Thurston counting and normal-surface Q-theory. It does not establish a general quasi-polynomial unknot recognizer: candidate generation, certified diagram-to-exterior provenance, and the accelerated hierarchy remain separate obligations.

Scope of the proved result. Given a validated finite triangulation with one torus boundary and an admissible binary normal vector, the delivered procedure counts its compressing-disc components in polynomial time in the encoded input. It can certify a positive count even when the surface is disconnected and its total boundary class is zero. A zero count concerns that vector alone.

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1 Result, scope, and relation to the repository

1.1 The concrete problem addressed

Let $\mathcal{T}$ be a finite triangulation of a compact connected orientable three-manifold whose boundary is one torus. Let $x \in \mathbb{Z}_{\geq 0}^{7t}$ be an admissible normal vector, where t is the number of tetrahedra. Write $S(x)$ for the corresponding embedded normal surface. The input coordinates are binary integers; neither $S(x)$ nor its components are supplied as expanded cell complexes. Define

$$\mathcal{D}(x) = \#\{C \in \pi_0(S(x)) : C \text{ is a disc with essential boundary}\}.$$

All multiplicities in this definition count actual components. In particular, 10^{100} parallel meridian discs contribute 10^{100} , not one.

The previous native backend obtains the number of components, orientability counts, boundary-curve count, and total Euler characteristic through interval orbits, with optional additional classification of closed and boundary-touching components. Its positive disc predicate requires the *whole supplied surface* to be connected. The maintained boundary-classification discussion explicitly leaves componentwise disc identification unresolved [12]. The new operation computes $\mathcal{D}(x)$ for a disconnected surface and supplies replayable evidence for the answer.

This is a sharply delimited algorithmic improvement. A positive answer proves that the supplied manifold has compressible boundary. If that manifold has independently been identified with the exterior of an input knot in S^3 , it certifies unknottedness. A zero answer only eliminates this particular vector. It does not eliminate other normal surfaces, prove boundary incompressibility, or establish knottedness. The delivered code does not silently add any of these missing implications.

1.2 Main statements

Theorem 1.1 (Three-weight disc query). *For the validated input class above, there is an algorithm that computes $\mathcal{D}(x)$ and a compressed histogram of componentwise Euler characteristic and boundary homology using three integer weights per orbit representative. Its running time, output length, and certificate length are polynomial in t and the maximum input-coordinate bit length. Its certificate can be checked without invoking its orbit-discovery scheduler or its weight-transport implementation.*

The topology in this theorem is proved in section 3; the compressed algorithm and its bit accounting are established in section 2. The checker shares the finite-manifold input validator and the deterministic cell-weight construction. Independence therefore concerns discovery and arithmetic transport, not two unrelated implementations of the entire topological input model. A further refinement to two weights after certified ambient homology preprocessing is proved in section 3.6. That variant is not implemented or included in the reported timings.

Theorem 1.2 (Primitive quadrilateral core). *For each global vertex v , let ℓ_v be the least triangle coordinate among the tetrahedral corners at v , and let L_v be its vertex-link vector. If x has a nonzero quadrilateral coordinate, let g be the gcd of its quadrilateral coordinates and set*

$$P = \frac{x - \sum_v \ell_v L_v}{g}, \quad Q = \max_j \frac{x_{q_j}}{g}.$$

Then P is an admissible nonnegative integral normal vector and

$$\mathcal{D}(x) = g\mathcal{D}(P).$$

Every coordinate of P is at most $\max(1, 4t - 1)Q$. If all quadrilateral coordinates vanish, $\mathcal{D}(x) = 0$ and no orbit computation is needed.

The full proof, including the effect of one-sided components under scaling, appears in section 4. The central computational consequence is that the expensive orbit query can use

$$O(\log(t + 1) + \log(Q + 1))$$

bits per geometric coordinate. Large common multiplicity and large independent vertex-link offsets still have to be read and processed, but they disappear from the subsequent orbit search. This remains useful when the gcd of the *original seven-coordinate vector* is one.

1.3 Which parts are established antecedents?

The weighted orbit method and polynomial component recovery are classical Agol–Hass–Thurston results. Canonical removal of vertex links and conversion between quadrilateral and standard coordinates are classical Q-theory. We use them openly rather than describe them as new topology [1, 2, 3]. The contribution here is the implemented specialization, its source-bound certificate, its exact disc-count reduction, and its measured improvement against more expensive routes to the same query. We do not claim that the specialized mathematical statements have no earlier appearance in the literature.

Ingredient	Status in this continuation
Weighted interval orbits	Classical algorithm; newly implemented as a checked extension of the maintained repository trace.
Vertex-link canonicalization	Classical construction, used with explicit binary-cost and quadrilateral-content accounting.
Three statistics for disc detection	A proved specialization that avoids recovering $7t$ weights or counting orientation and boundary covers.
Two weights with ambient homology	Proved refinement; implementation and performance comparison remain future work.
Disconnected disc census	New native repository capability with finite oracles and source-bound replay.
General quasi-polynomial recognition	Remains unimplemented and unproved for the delivered pipeline.

1.4 Why aggregate invariants are insufficient

There are two different pitfalls. First, even parallel meridian discs have zero *total* mod-two boundary class. Thus requiring a nonzero total class rejects valid positive examples. Second, a surface can have total Euler characteristic one and a nonzero total boundary class without containing a disc. For example, a small interior vertex-link sphere disjoint from an orientable genus-one spanning surface has total Euler characteristic $2 + (-1) = 1$. The sphere has no boundary and the genus-one component is not a disc. The corresponding finite normal-surface example is included in the external audit.

The new query associates Euler characteristic and boundary class to the *same connected component*. Its correctness does not follow from a test of the sums of these quantities. This distinction is the main reason weighted orbits are needed.

1.5 Exact API contracts

The new APIs are additive modules under **fastunknot**; existing recognizer defaults, certificate versions, and old aggregate APIs are unchanged. The following table specifies the information computed by each route.

Route	Dimension	Certified output
<code>mode='disk'</code>	3	Component histogram of $(\chi, h_1, h_2 \bmod 2)$, total components, and essential-disc count.
<code>mode='summary'</code>	5	The preceding data, normal-disc count, and boundary intersection weight for each histogram type.
<code>mode='coordinates'</code>	$7t$	Actual connected-component normal vectors, grouped with binary multiplicities, and their disc predicates.
Reduced count API	3 on P	Essential-disc count of the original x , with a certified canonical reduction.

Boundary intersection weight is the number of normal boundary edge points, equal to the number of boundary normal arcs. It is *not* the number of boundary circles. The reduced count API intentionally makes no claim about the total number of original components: that quantity is not homogeneous in the presence of one-sided components. The full-coordinate route can extract a concrete disc vector for a later cutting operation, but finding and certifying that cut is outside this package.

1.6 The quasi-polynomial target in the inspected sources

The explicit March 2021 slides announce $2^{O((\log n)^3)} = n^{O((\log n)^2)}$ time; the institutional announcement instead uses $n^{c \log n}$. Both are quasi-polynomial, but the exponents are different [6, 7]. The July 2026 preprint provides a hierarchy algorithm and polynomial verification results, while separating iteration counts from step costs that have not all been specified sufficiently for runtime analysis [8]. The repository makes the same distinction. Consequently this article treats quasi-polynomial time as the global implementation target and states the local proved bounds explicitly.

The selected source snapshot is commit 7518823550fb, whose full identity is recorded in the provenance manifest. This pins the comparison despite active development elsewhere in the repository. No remote repository changes are required to inspect, reproduce, or integrate the accompanying archive.

2 Weighted interval orbits with polynomial output

Weighted orbit counting is classical. Agol, Hass, and Thurston prove a polynomial-time weighted extension of their interval algorithm in Theorem 16; their Corollary 17 applies it to the normal coordinates of connected surface components [1]. We use this antecedent explicitly. The contribution here is a maintained implementation with a precise suffix-transport invariant, a compressed histogram contract, independently checked arithmetic, and the small set of weights needed by the subsequent disc query. Neither weighted counting itself nor the deduction that normal components can be recovered in polynomial time is claimed as new.

2.1 Input, output, and the surviving-representative invariant

Let $X = \{0, \dots, N-1\}$, with an equivalence relation generated by K interval pairings. Each pairing is an isometry between equal-length integer intervals, with either sign. Endpoints and N are binary integers. A weight function $w : X \rightarrow \mathbb{Z}^d$ is supplied by s additive half-open intervals:

$$w(x) = \sum_{i=1}^s v_i \mathbf{1}_{[\ell_i, r_i)}(x), \quad v_i \in \mathbb{Z}^d.$$

Intervals may overlap, coordinates may be negative, and uncovered points have zero weight. Sorting their endpoints and summing differences produces a canonical list of constant runs covering X . Adjacent equal runs are merged. For $N > 0$ the initial number R_0 of runs is at most $2s + 1$; the empty universe has no runs.

For an orbit O , define $W(O) = \sum_{x \in O} w(x)$. The output is the sorted histogram

$$H(v) = \#\{O : W(O) = v\}, \quad H(v) > 0$$

on its explicitly represented support. Multiplicities remain binary integers. The algorithm does not emit one record per orbit. For example, an unpaired universe of size 2^b with constant weight v has the one-record answer $H(v) = 2^b$. Expanding that answer into a list would destroy the claimed encoded complexity. Distinct components with the same weight deliberately share a record; their identities and attaching maps are not part of this contract.

The unweighted producer supplies a complete AHT reduction trace. It contains identity deletion, reflection trimming, periodic merging, transmission, static-interval contraction, and suffix truncation. Both the classical and Fine–Wilf merger versions are supported by the existing independent trace checker. Weight propagation reconstructs the current pairing list from these events, without running the discovery scheduler again.

The relevant invariant is stronger than conservation of the total weight. Each surviving point represents a disjoint collection of original points; its current weight is their vector sum. The current orbit relation groups these collections into exactly the original orbits not yet emitted. Identity deletion, trimming, merging, and transmission change the generators of the relation without removing points. They therefore leave all point weights unchanged. Only the two point-removal operations require weight updates.

2.2 Exact transport through a truncation

Suppose the current universe is $[0, N)$ and a checked event removes the suffix $[c, N)$. That suffix is incident only to its designated carrier pairing. For a translation carrier with displacement $p > 0$, its domain begins at a , its range begins at $a + p$, and $c \geq a + p$. Repeated inverse application sends every deleted point to the retained interval $J = [c - p, c)$ by

$$f(x) = c - p + ((x - (c - p)) \bmod p). \quad (1)$$

All intermediate inverse applications are defined: until the first value below c , the point remains in the carrier's range. The target belongs to $[a, c)$ and is retained. Each original orbit still has a representative.

Consider one constant suffix run $[\ell, r)$ of value v . Write

$$r - \ell = qp + u, \quad 0 \leq u < p, \quad z = c - p + ((\ell - (c - p)) \bmod p).$$

Its pushforward contributes qv to every point of J . The remaining u points contribute v on the cyclic interval of length u starting at z . Putting $u_1 = \min(u, c - z)$, this is the sum of contributions on

$$[z, z + u_1), \quad [c - p, c - p + u - u_1).$$

Empty intervals are omitted. Thus a run of astronomical length needs one division with remainder and at most three constant-interval additions. The retained prefix keeps its existing weights, and all suffix contributions are added to it using an endpoint-difference table.

A checked reflection carrier has disjoint domain and range after trimming. With current right endpoint $N - 1$ and left endpoint a , its inverse is $x \mapsto a + N - 1 - x$. Consequently a deleted run $[\ell, r)$ maps bijectively to

$$[a + N - r, a + N - \ell).$$

This interval lies wholly in the retained prefix. The implementation adds the same vector there and discards the source suffix. The reflection's order reversal must be retained; treating every pairing as a translation would give incorrect component sums even when the unweighted count agreed.

For a static contraction, every point in each supplied gap is a singleton orbit of the current relation. Its accumulated weight is already the full weight of one original orbit. Intersecting the gaps with the current runs therefore adds the intersection lengths to the corresponding histogram entries. The remaining points and weights are shifted by the same order-preserving gap-removal map used for the pairings.

Proposition 2.1 (Correct weighted replay). *Replaying these operations along a valid complete AHT trace returns exactly the histogram of original orbit-weight sums, including signed weights.*

Proof. Maintain the disjoint representative collections described above. The four relation-only operations preserve them unchanged. A suffix truncation assigns each deleted collection to its image under the carrier's inverse, possibly iterated. These assignments partition all deleted collections and preserve their original orbit membership. Summing their vectors gives exactly the displayed pushforward formulas. A static point then represents one complete original orbit and can be emitted. The checked trace finishes with no points and no pairings, so every original orbit is emitted exactly once. No step uses positivity; the proof applies coordinatewise in the additive group $\mathbb{Z}^d$. $\square$

2.3 A bound on run growth and histogram size

The fact that one source run gives constantly many intervals is insufficient to prove polynomial space: repeated constant-factor growth could still be exponential. The necessary bound follows from transporting jumps.

Lemma 2.2 (Additive run growth). *A truncation increases the number of canonical weight runs by at most two. A static contraction does not increase it. Thus after E trace events,*

$$R \leq R_* := R_0 + 2E.$$

Proof. Split the old runs at the truncation point into P nonempty prefix runs and Q nonempty suffix runs. At most one old run was split, so $P + Q \leq R + 1$. Extend the suffix function by zero to the integer line. Its discrete derivative has jumps only at its $Q + 1$ endpoints. Under translation folding, each jump has one residue image modulo p . The cutoff endpoint maps to the wrap boundary $c - p$, leaving at most Q other interior residue locations. Relative to the $P - 1$ internal prefix boundaries, these locations and the possible new wrap boundary give at most $P + Q + 1$ runs. This is at most $R + 2$. Coincident endpoints and cancelling jumps only reduce the count.

For a reflection, the $Q + 1$ suffix endpoints have one reflected image each. Together with the $P - 1$ prefix boundaries they again give at most $P + Q + 1 \leq R + 2$ runs. Finally, deleting static intervals retains a subsequence of the old constant runs. Every new change of value corresponds to at least one old change between consecutive retained points. Coalescing adjacent equal values therefore cannot increase the run count. $\square$

If a contraction contains g gaps, their intersections with a run list have at most $R + g - 1$ nonempty pieces. There are at most $2K + 1$ gaps, since all pairing domains and ranges together have at most $2K$ occupied intervals. Pairing count never increases during the trace. Consequently the number h of distinct histogram records satisfies the conservative bound

$$h \leq E(R_0 + 2E + 2K + 1). \quad (2)$$

Repeated equal weights share a record, so this bounds the actual output as well as the number of possible emissions. Since the maintained AHT trace length is polynomial in K and $\log(N+1)$, the histogram has polynomial encoded size. This observation is essential when the number of components itself is exponentially large.

2.4 Arithmetic and certificate complexity

Assume every coordinate of every supplied v_i has absolute value at most A . Each current representative contains a subset of original points, hence every accumulated coordinate has magnitude at most NsA . This also bounds each completed orbit sum. Taking

$$B = O(1 + \log(N+1) + \log(s+1) + \log(A+1))$$

therefore bounds all weight bit lengths. Multiplicities use $O(1 + \log(N+1))$ bits. Quotient multiplication is integer multiplication on these bit lengths, not a unit-cost operation on arbitrarily large numbers.

Initial normalization uses $O(sd)$ coordinate additions and $O(s \log(s+1))$ endpoint comparisons. Given a trace of E events, one truncation uses $O(dR_*)$ coordinate operations and $O(R_* \log(R_*+1))$ endpoint comparisons. A contraction requires at most $O(d(R_*+K) + (R_*+K) \log(R_*+K+1))$ such operations. Source reconstruction and the final histogram sort add polynomial costs; lexicographic sorting of h vectors uses at most $O(dh \log(h+1))$ coordinate comparisons. A useful event-sensitive accounting bound, apart from the existing unweighted producer and checker, is therefore

$$O\left(sd + s \log(s+1) + E[d(R_*+K) + (R_*+K) \log(R_*+K+1) + K^2] + dh \log(h+1)\right).$$

The K^2 term charges the independent weighted checker's direct reconstruction of contracted pairing endpoints across up to $O(K)$ gaps. The producer can use the sharper bound without that term. This counts arithmetic and ordinary dictionary operations. Integer hashing, comparisons, additions, and products must then be charged their bit costs. Adversarial collisions in Python dictionaries can worsen the displayed container-operation model, but still give a polynomial bound for these explicit polynomial-size tables; deterministic balanced maps give the usual additional logarithmic alternative. No conclusion depends on treating a 2^b -bit integer as one machine word.

The certificate stores the canonical source weights, the existing unweighted orbit proof, and the claimed sorted histogram. A contraction event can list $O(K)$ gaps, so trace payload size is charged as well as event count. The run bound, output bound, and bit bound together show polynomial certificate length. Internally the producer creates an orbit trace even when the caller does not request a returned certificate: weight propagation needs that sequence of contractions. A supplied trace can instead be source-checked and reused, avoiding a new orbit search.

2.5 Independent arithmetic and sufficient statistics

The independent weighted checker first verifies all unweighted local rules against the caller's pairing list. It then reconstructs the pairing updates and weight transport separately. In particular, its translation arithmetic does not reuse the producer's quotient-and-remainder interval additions. For the zero-extended suffix function v and a target representative y , put

$$S(y) = \sum_{k \in \mathbb{Z}} v(y + kp).$$

At the leftmost residue $y = c - p$, a source run $[\ell, r)$ contributes its weight multiplied by

$$\left\lfloor \frac{r - 1 - y}{p} \right\rfloor - \left\lfloor \frac{\ell - 1 - y}{p} \right\rfloor.$$

This determines one anchor value. The identity

$$S(y) - S(y - 1) = \sum_{k \in \mathbb{Z}} (v(y + kp) - v(y - 1 + kp))$$

determines every remaining value by integrating the independently mapped suffix jumps. These two arithmetic constructions reach the same required pushforward through different calculations. The checker additionally checks the total orbit count, exact sorted histogram, source binding, and signed mass conservation. A partial histogram is never returned when the underlying orbit allowance is exhausted; cancellation callbacks propagate through construction and replay.

Finally, many decisions depend on a small additive signature rather than a full component description. If an orbit predicate factors through d_0 additive statistics, those statistics may be transported directly and the desired count obtained by summing histogram multiplicities satisfying the predicate. The number of vector coordinates in the arithmetic bound becomes d_0 . In the next section, three statistics suffice for the disc predicate, whereas full normal-component extraction uses $7t$ coordinates. This removes a factor of order t from the dimension-dependent arithmetic bound; it does not remove common orbit-search and endpoint-processing costs. The signatures also need not recover the normal vector of a particular disc. A subsequent operation that actually cuts along a selected component can request the full-coordinate mode and pay for its richer output.

3 Cell weights and three-statistic disc detection

3.1 Validated finite geometry

The existing validator checks reciprocal face pairings, face-connectedness, ambient orientability, absence of reversed edge identifications, spherical or disc vertex links, and a single finite boundary torus. Ideal vertices and other boundary topologies are rejected. Normal coordinates are checked for nonnegativity, face matching, and the quadrilateral constraints. These checks are retained before any new component query.

The proof requires no irreducibility hypothesis on the supplied manifold. Irreducibility becomes relevant to other recognition and hierarchy operations, not to the classification of its embedded surface components. In particular, the implementation must correctly reject a closed projective-plane component as a disc even though its Euler characteristic is one.

For every global triangulation edge e , let w_e be its normal intersection number. A block of w_e consecutive integers represents those intersection points, ordered using a consistent orientation of that edge. Concatenate these blocks to obtain a universe of size

$$N = \sum_e w_e.$$

Each normal arc in a face pairs its two endpoint positions. Consecutive arcs of a fixed type give one interval isometry. Count a glued face once and each boundary face once. The number of pairings is at most $12t$. The graph formed by these vertices and arc edges is the one-skeleton of the normal surface's polygonal cellulation. Its connected components are exactly those of the surface. Identifications within one tetrahedron, loop arcs, and multiple edges cause no problem for either connectivity or the Euler formula.

3.2 Integral ownership of Euler characteristic

One can distribute fractional Euler weights over polygon corners. We instead use an integral ownership rule, which avoids denominators entirely. Give every edge-intersection point an initial Euler weight $+1$. For every normal arc, charge -1 to one of its endpoints. For every normal polygon, charge $+1$ to one selected corner. The selected corner belongs to the same surface component as the entire polygon; likewise the endpoint of an arc. Therefore, for every component C ,

$$\sum_{p \in C} w_\chi(p) = V(C) - E(C) + F(C) = \chi(C).$$

This is true even when several selected local corners become the same global intersection point: their additive contributions must then accumulate there.

All ownership data is described by $O(t)$ intervals. The negative arc charge is assigned on the canonical source interval of each maintained pairing. For a triangle type at vertex v , choose a fixed incident local edge and use the triangle points nearest v . For an active quadrilateral type, choose the lexicographically first local edge that it crosses. Along an edge from local vertex a to local vertex b , the points appear as triangles at a , then quadrilaterals, then triangles at b . If their relevant counts are T_a, Q , the selected quadrilateral corners occupy

$$[T_a, T_a + Q).$$

Reflect this interval if the local edge direction disagrees with the global orientation. No stack is expanded.

Lemma 3.1 (Cell-weight identity). *The constructed piecewise-constant integer weight has component sums $\chi(C)$, and has an $O(t)$ -interval additive description. A second weight obtained by retaining just the polygon credits sums to the number of normal polygons in C .*

Proof. Every cell receives exactly one charge of its appropriate sign and each charge lies in its own connected component. A tetrahedron has at most five nonempty polygon types, and a face has at most three arc types. The listed charges are constant on each type's consecutive set of corners or endpoints. $\square$

The full-coordinate reference replaces each polygon credit by the corresponding standard basis vector of $\mathbb{Z}^{7t}$. A component's orbit sum is then its actual normal coordinate vector. This is the classical component-recovery route [1]; the new disc query avoids its growing vector dimension.

3.3 A small, certified homology basis of the boundary

Let G be the primal one-skeleton of the boundary triangulation. Self-loops and multiple edges are permitted. Select a spanning tree T of G . In the dual graph select a spanning tree T^* using edges whose primal edges are outside T . Exactly two primal edges are in neither tree, since the boundary is a torus and

$$|E| - (|V| - 1) - (|F| - 1) = 2.$$

For each leftover edge, append the unique path in T connecting its endpoints. These two fundamental cycles, c_1, c_2 , form a basis of $H_1(\partial\mathcal{T}; \mathbb{F}_2)$.

For completeness, the tree-cotree fact can be seen by contracting the primal tree and removing edges dual to a dual spanning tree. These operations preserve the surface and reduce its cellular description to one vertex, one face, and two edges. Over $\mathbb{F}_2$, the boundary of that face is zero, and the two surviving cycles freely generate the two-dimensional first homology.

Reversing the contractions gives exactly the chosen fundamental cycles. The dual graph outside the primal tree is connected: a separating dual cut would correspond to a nonempty primal cycle contained in the tree.

The certificate checker does not rely on the spanning-tree algorithm's output being correct. It independently checks that each proposed cycle has even incidence at every vertex, constructs the boundary vectors of all triangles, and performs binary Gaussian elimination. The two proposed cycles must each increase the rank modulo the face-boundary span. Since the validator has already established a torus, two independent classes are a basis. Repeated edges of a triangular face cancel modulo two in this check.

3.4 Boundary intersection weights

For each $j \in \{1, 2\}$, give every intersection point on a boundary edge in c_j a homology weight $+1$ in coordinate j ; give every other point weight zero. If $z_e(C)$ counts the boundary points of component C on edge e , then its orbit sum is

$$\eta_j(C) = \sum_{e \in c_j} z_e(C).$$

The code retains this as an exact integer and reduces it modulo two when interpreting the result. Equivalently, one could carry an $\mathbb{F}_2$ coordinate throughout the weighted algorithm; the present implementation uses one uniform integer-vector engine for all query modes.

The parity vector $z_e(C) \bmod 2$ is the cellular cocycle Poincaré-dual to ∂C : every boundary triangle meets the normal boundary curve in arcs and therefore has even total endpoint incidence. Evaluating that cocycle on the two basis cycles yields

$$h(C) = (\eta_1(C), \eta_2(C)) \bmod 2, \quad h(C) = 0 \iff [\partial C] = 0 \text{ in } H_1(\partial\mathcal{T}; \mathbb{F}_2).$$

The equivalence uses the nondegenerate homology-cohomology pairing over the field $\mathbb{F}_2$. It applies to the total boundary of each component, rather than to one arbitrarily selected boundary circle.

3.5 Three additive weights characterize compressing discs exactly

Let C be a connected component of the supplied proper normal surface. The cell-incidence weight gives $\chi(C)$. Let γ_1, γ_2 be the checked homology basis of the boundary torus described above, and define

$$h_i(C) = \sum_{e \in \gamma_i} |C \cap e| \pmod{2}, \quad i = 1, 2.$$

The raw orbit calculation retains the corresponding integer sums; parity is taken only when interpreting the resulting component signature. In particular, signed Euler arithmetic and binary homology arithmetic are not silently identified.

Lemma 3.2. *The pair $(h_1(C), h_2(C))$ vanishes exactly when the mod-two boundary class of C vanishes in the ambient torus.*

Proof. On every boundary triangle, a normal arc contributes two endpoints. Consequently the parity of edge intersections is a cellular 1-cocycle, including in generalized triangulations where repeated face edges must be combined modulo two. This cocycle represents the Poincaré dual of the boundary multicurve. Evaluation on a homology basis determines its cohomology class, since evaluation gives the nondegenerate pairing between $H^1(T^2; \mathbb{F}_2)$ and $H_1(T^2; \mathbb{F}_2)$. Both spaces have dimension two. Thus the two values vanish precisely for the zero boundary class. $\square$

Theorem 3.3 (Exact three-weight disc criterion). *A connected component C is a compressing disc for the ambient boundary torus if and only if*

$$\chi(C) = 1, \quad (h_1(C), h_2(C)) \neq (0, 0).$$

Proof. A compressing disc has Euler characteristic one. Its boundary is an embedded essential circle on the torus. Such a circle represents a primitive integral slope, so its reduction modulo two is nonzero. The lemma proves necessity.

Conversely, the nonzero boundary class implies that C has boundary. The classification of connected compact surfaces now leaves only a disc. Indeed, an orientable surface of genus a with $b \geq 1$ boundary circles has $\chi = 2 - 2a - b$; equality to one forces $a = 0, b = 1$. A nonorientable surface with $k \geq 1$ crosscaps and $b \geq 1$ boundary circles has $\chi = 2 - k - b \leq 0$. The resulting disc has nonzero boundary class, and hence its simple boundary circle is essential on the torus. $\square$

The boundary condition in this argument is necessary. A closed projective plane also has Euler characteristic one; its two boundary evaluations are zero, so it is correctly excluded. Such closed one-sided surfaces occur in the supported general torus-boundary manifold fixtures, even though those fixtures need not be knot exteriors. No assumption of irreducibility is smuggled into the local component test.

Two binary linear probes are also the minimum needed to recognize the zero element of the full torus boundary-class space by linear evaluation alone. A single linear functional on the two-dimensional space has a nonzero kernel. The two-probe statement concerns this unrestricted homology interface; it does not prove a lower bound for a specially restricted collection of normal surfaces in one fixed manifold, nor for nonlinear preprocessing. Together with the Euler coordinate, it explains the natural three-weight implementation and its reduction from $7t$ -dimensional coordinate extraction.

The torus hypothesis cannot be dropped from the equivalence. On a boundary surface of genus at least two, an essential separating simple circle can have zero mod-two homology. The same predicate would then give only a sufficient test for certain compressing discs, and could miss others. The native input validator's one-torus-boundary contract is therefore part of the theorem, not merely a convenience for constructing a small basis.

Finally, the test must be applied to individual component weights. Two parallel meridians have zero total boundary class but contribute two compressing discs. Conversely, an inessential vertex-link disc together with a Möbius band can have total Euler characteristic one and nonzero total boundary class while contributing no compressing disc. A weighted orbit histogram resolves both ambiguities without listing the components one by one.

3.6 A proved two-weight refinement using ambient homology

The three-weight implementation uses only the boundary torus when choosing its homology probes. The ambient manifold supplies additional information: boundaries of proper surfaces occupy a one-dimensional subspace of the torus homology. Exploiting that restriction gives the following proved refinement. It is *not implemented or benchmarked in this delivery*; the reported disc-query improvements use the implemented three-weight mode and the stated comparison arms.

Proposition 3.4 (One ambient-certified boundary probe). *Let M be a compact connected orientable three-manifold with exactly one torus boundary, and let $i : \partial M \hookrightarrow M$. There exists a boundary cycle γ with $i_*[\gamma] \neq 0$ in $H_1(M; \mathbb{F}_2)$. For any such cycle and every connected proper normal-surface component C ,*

$$C \text{ is a compressing disc} \iff \chi(C) = 1 \text{ and } |\partial C \cap \gamma| \equiv 1 \pmod{2}.$$

The intersection count can be computed by the same additive edge-point weights as before. Thus two integer weights suffice after the indicated ambient preprocessing.

Proof. All homology groups in this proof have coefficients in $\mathbb{F}_2$. Write $b_j = \dim H_j(M)$. Poincaré–Lefschetz duality gives $\chi(M, \partial M) = -\chi(M)$; combined with the Euler characteristic of the pair, this gives $2\chi(M) = \chi(\partial M) = 0$. Since M is connected with nonempty boundary, $b_0 = 1, b_3 = 0$, and hence $b_1 - b_2 = 1$ [11].

In the long exact sequence of the pair, $H_0(\partial M) \rightarrow H_0(M)$ is an isomorphism. Consequently $H_1(M) \rightarrow H_1(M, \partial M)$ is surjective. Duality identifies the dimension of the latter group with b_2 . Exactness therefore gives $\dim \operatorname{im} i_* = b_1 - b_2 = 1$. The kernel

$$L = \ker(H_1(\partial M) \longrightarrow H_1(M))$$

is a line in the two-dimensional torus homology. Every compact surface, including a nonorientable one, has a relative fundamental class over $\mathbb{F}_2$. Pushing that class into $(M, \partial M)$ shows that $[\partial C] \in L$.

The torus intersection pairing is nondegenerate and alternating. A line L consequently has orthogonal complement L itself. Since $i_*[\gamma] \neq 0$, the class $[\gamma]$ is outside L and pairs to one with its unique nonzero element. Odd intersection with γ is therefore equivalent to $[\partial C] \neq 0$. The surface-classification and primitive torus-slope argument already proved then gives the disc criterion. This argument imposes neither irreducibility nor a torsion-free integral homology assumption. $\square$

There is a concrete finite preprocessing algorithm. Embed the two verified boundary basis cycles into the ambient global edge-chain space over $\mathbb{F}_2$. Form the span of boundaries of all ambient triangular face classes, counting paired faces once and combining repeated edges modulo two. At least one boundary basis cycle lies outside this span, because the boundary inclusion has rank one. Choose such a cycle as γ . It is already a cycle, so being outside the face-boundary span is exactly the required nonzero ambient homology condition.

The preprocessing can carry a particularly small certificate: an ambient cellular 1-cocycle u satisfying

$$u(\partial f) = 0 \text{ for every triangular face } f, \quad u(\gamma) = 1.$$

These identities prove that γ is not a boundary. A producer can find u by Gaussian elimination in $O(t^3)$ elementary operations over $\mathbb{F}_2$; replay checks $O(t)$ constant-size face equations and the cycle evaluation, without repeating elimination. The cycle and cocycle depend only on the triangulation and can be reused across its candidate surfaces.

Both signatures have constant dimension. This refinement saves one transported integer coordinate while adding ambient preprocessing; its practical benefit remains to be measured. It does not contradict the earlier observation about two linear probes on the *unrestricted* torus class space. The new probe uses a certified geometric restriction to the line L . With several boundary components the rank argument above changes, so this particular one-probe conclusion must not be transferred to that setting without a new analysis.

3.7 Component histograms and witness extraction

In disc mode the weighted output is initially a histogram of exact integer triples. The result interpreter reduces its last two entries modulo two and merges identical triples, adding their binary multiplicities. This last aggregation is safe because the predicate depends only on those three displayed values. It is unnecessary to distinguish normally nonisotopic components that have the same disc-test signature.

Summary mode adds the number of polygons and the number of boundary intersection points. The latter equals the number of boundary normal arcs componentwise, since every boundary circle has an equally sized vertex and edge set in its induced cellulation. These two counts are diagnostic data; they do not determine the number of boundary circles or the full genus.

When an actual disc vector is required, the full-coordinate mode recovers a histogram of connected-component coordinate vectors. One may select any record with the certified disc predicate. The coordinate vector is explicit in $7t$ binary entries, while a large number of identical copies is retained as a multiplicity. The existing unweighted trace can be checked and reused, so extraction does not require another search for an orbit-reduction order. This separates a cheap decision query from a more informative extraction query without conflating their costs.

3.8 Why the geometric restriction matters

The boundary-torus assumption is substantive. On a boundary surface of genus at least two, an essential separating simple closed curve has zero mod-two homology. Such a curve could bound a compressing disc and would escape the three-weight test. A higher-genus extension therefore needs compressed curve-essentiality information, not just more copies of a homology basis. The supplied validator rejects these inputs, so the implementation does not silently apply the torus theorem outside its hypotheses.

4 A canonical quadrilateral kernel for exact disc counts

The weighted census answers a stronger question than an aggregate topology summary, but it still receives endpoints whose bit lengths reflect every normal-coordinate multiplicity. The maintained common-gcd reduction does not remove a large additive collection of vertex-link surfaces. Such additions can make the gcd of all $7t$ coordinates equal to one while almost every coordinate has thousands of bits. For the particular question of counting essential disc components, we can remove both sources of numerical growth.

The coordinate canonicalization is classical Q-theory. Tollefson introduced quadrilateral coordinates, and Burton describes equal quadrilateral data, vertex-link ambiguity, canonical representatives, positive homogeneity and minimum-triangle subtraction explicitly [2, 3]. We use these facts with proofs because their exact hypotheses and their connection to the maintained interval interface matter here. The delivered specialization is a source-bound essential-disc count, together with its parameter-sensitive query bound and replayable arithmetic reduction.

4.1 Remove precisely the normal vertex-link components

Let x be an admissible normal vector in the supported finite triangulation with t tetrahedra. For a global vertex v , let $\mathcal{C}_v$ be the set of tetrahedron corners identified with v , and write $n_v = |\mathcal{C}_v|$. The sets $\mathcal{C}_v$ partition the $4t$ corners. Write $T_c(x)$ for the triangle coordinate at corner c , and let L_v be the vector with one triangle at every corner in $\mathcal{C}_v$ and zero elsewhere. It represents the vertex link: a sphere at an interior vertex and a disc at a boundary vertex.

Define

$$\ell_v = \min_{c \in \mathcal{C}_v} T_c(x), \quad Y = x - \sum_v \ell_v L_v.$$

Thus Y has a zero triangle coordinate at some corner of every global vertex. Its quadrilateral coordinates equal those of x .

Lemma 4.1 (Geometric meaning of vertex-link subtraction). *The vector Y is admissible, and the embedded normal surface represented by x is normally isotopic to the disjoint union of the surface represented by Y and ℓ_v copies of the vertex link at every v .*

Proof. Nonnegativity follows from the definition of each minimum, and no quadrilateral coordinate changes. A matching equation across a face contains one triangle coordinate from each of two corners identified with the same global vertex. Subtracting ℓ_v from both preserves that equation.

Realize the admissible vector Y as a normal surface. Since the surface is finite and misses the triangulation vertices, sufficiently small vertex neighborhoods are disjoint from it. Nested frontiers of these neighborhoods give the requested vertex-link copies, also disjoint from one another. Their disjoint union with the realization of Y has coordinate vector x . Uniqueness of a normal realization with its admissible coordinates, up to normal isotopy, gives the asserted decomposition. This argument uses an actual disjoint union; it does not assume an arbitrary normal sum preserves its summands as components. $\square$

Every boundary vertex-link disc has contractible boundary on the boundary torus. Its presence therefore increases the number of discs without increasing the number of *compressing* discs. Interior vertex links are closed and contribute no such disc either. These facts justify removing the links before the specialized query, even when their multiplicities dominate the input size.

4.2 Quadrilateral content is the full content of the peeled vector

Suppose at least one quadrilateral coordinate is nonzero, and let

$$g = \gcd\{Q_{\tau,j}(x) : 1 \leq \tau \leq t, 1 \leq j \leq 3\}.$$

Zero entries participate in the ordinary gcd convention. Let $Q_{\max}$ be the largest quadrilateral coordinate and set $r = Q_{\max}/g$. In particular $g \geq 1$ and $r \geq 1$.

Theorem 4.2 (Canonical primitive quadrilateral core). *Every coordinate of Y is divisible by g , and its full coordinate gcd is exactly g . The vector $P = Y/g$ is admissible, primitive and has a zero triangle at every global vertex. Moreover,*

$$T_c(P) \leq (n_v - 1)r \quad (c \in \mathcal{C}_v), \quad \max_i P_i \leq (4t - 1)r.$$

The vector P is uniquely determined by the primitive quadrilateral vector $Q(x)/g$. If all quadrilateral coordinates of x are zero, then $Y = 0$.

Proof. Consider the graph whose vertices are the corners in $\mathcal{C}_v$, with edges recording their identifications across paired faces. It is connected because those face identifications define the global vertex class. Along each graph edge, the normal matching equation has the form

$$T_c(Y) - T_{c'}(Y) = Q' - Q,$$

where Q, Q' are individual quadrilateral coordinates, possibly zero. The right side is divisible by g . Choose a corner with triangle coordinate zero and follow a graph path to any other

corner. Its triangle coordinate is divisible by g as well. All quadrilateral coordinates are already divisible by g . Conversely, a common divisor of all coordinates must divide their quadrilateral subset, whose gcd is g . The full gcd is therefore exactly g .

After division, every quadrilateral appearing in one matching equation lies between zero and r , so each triangle difference has absolute value at most r . A simple path from a zero corner has at most $n_v - 1$ edges. Summing the differences proves the triangle bound. The global bound follows from $n_v \leq 4t$ and the bound r on quadrilateral coordinates.

If two vectors have the same quadrilateral coordinates, the difference between their triangles is constant on each connected corner graph. Requiring minimum zero at each vertex fixes every such constant. This proves uniqueness of the vertex-reduced lift and hence of P . Finally, if every quadrilateral vanishes, all triangle differences on each corner graph vanish; its minimum zero forces every residual triangle to vanish. $\square$

The theorem is invoked only after validating an actual supplied standard normal vector. It does not assert that arbitrary quadrilateral data has a finite normal lift, and it does not replace the matching and quadrilateral constraints with an unchecked Q -coordinate input.

4.3 Essential-disc counts survive the nonlinear sheet behavior

Let $\mathcal{D}(z)$ denote the number of compressing-disc components in the embedded normal surface of vector z . The scalar behavior of this number is simpler than the scalar behavior of the full component count.

Lemma 4.3 (Homogeneity of the essential-disc count). *For every admissible normal vector z in the supported orientable manifold and every positive integer k ,*

$$\mathcal{D}(kz) = k\mathcal{D}(z).$$

Proof. Realize the surface with coordinates kz in the normal interval neighborhood of the surface with coordinates z . Local normal sheet orders are either preserved or reversed. A two-sided component therefore produces k disjoint copies. Each compressing disc is two-sided, and these copies have isotopic essential boundaries on the ambient torus.

A one-sided component has nontrivial normal monodromy: sheet j is paired with sheet $k - 1 - j$. For even k , this produces $k/2$ copies of its connected normal orientation double. For odd k , it produces $(k - 1)/2$ such doubles and one original one-sided component. The ambient manifold is orientable, so a one-sided surface is nonorientable. If it has boundary, its Euler characteristic is at most zero, and the orientation double also has Euler characteristic at most zero. Neither can be a disc. If the component is closed, every such cover remains closed. Thus one-sided components never create discs through this scaling process. Exactly the original compressing discs contribute, each with multiplicity k . $\square$

Corollary 4.4 (Exact count on the primitive core). *With the notation above, if $g > 0$ then*

$$\boxed{\mathcal{D}(x) = g\mathcal{D}(P).}$$

If all quadrilateral coordinates vanish, then $\mathcal{D}(x) = 0$ and no orbit query is needed.

Proof. Apply Lemma 4.1; none of the removed vertex links contributes an essential disc. Then apply Lemma 4.3 to $Y = gP$. In the triangle-only case the peeled vector is zero. $\square$

The count-only interface deliberately reports no full component count for x . For example, twice a normal Möbius band is an annulus, so multiplying its original component count by two is incorrect. Both surfaces have zero compressing-disc components, consistently with

the corollary. The separate unreduced census remains the appropriate interface for complete component histograms and coordinate extraction.

The corollary asserts scalar homogeneity after a specific geometric decomposition. It does not assert additivity of $\mathcal{D}$ under arbitrary compatible normal-vector addition: regular exchanges in a general normal sum can change the connected components. Vertex links are removable here because Lemma 4.1 realizes that particular addition as a disjoint union.

4.4 Input arithmetic and reduced query complexity

There are at most t nonzero quadrilateral coordinates by admissibility. The coefficient bound gives a convenient coarse disc-count estimate

$$W(P) = \sum_i P_i \leq (16t^2 - 3t)r \leq 16t^2r.$$

The maintained edge-point orbit universe has at most $4W(P)$ points, so its endpoints use

$$b_0 = O(\log(t+1) + \log(r+1))$$

bits. This bound is independent of g and the vertex-link multiplicities. Only one three-weight component query is required on P .

For an explicit accounting, let B bound the original coordinate bit lengths, let $V(t, B)$ denote full source validation and decoding, and let $\mathcal{P}(t, b)$ denote the polynomial cost of the delivered three-weight census on a source whose coordinates and endpoints have $O(b)$ bits. Writing $G(B)$ and $\text{Div}(B)$ for binary gcd and division costs, the count-only route has the bound

$$\begin{aligned} &V(t, B) + O(tB + tG(B) + t\text{Div}(B)) \\ &\quad + \mathcal{P}(t, O(\log(t+1) + \log(r+1))) + O(M(B + \log(t+1))), \end{aligned}$$

where M accounts for the final binary multiplication. Comparisons and subtractions for the vertex minima are charged in tB . A second validation of the small core, as performed by the current wrapper, is included in $\mathcal{P}$. In the triangle-only case the census term is absent.

The certificate binds the full original source, records the vertex-link multiplicities, the divisor, the explicit core and its independently checked three-weight proof, and verifies the final product. Thus the outer arithmetic fields still depend on the original input size. Large multiplicity is not removed from the cost of reading, validating or binding that input; it is removed from every endpoint and weight in the expensive core orbit trace.

Example 4.5 (An arbitrarily large source with full coordinate gcd one). Take a primitive, vertex-reduced meridian vector D in a one-vertex layered solid torus, with quadrilateral gcd one, and let L be the boundary vertex link. For any positive binary integer g , set

$$x_g = gD + (g+1)L.$$

The quadrilateral coordinates have gcd g , while a triangle at a zero corner of D equals $g+1$. Hence the gcd of all coordinates is one. Common multiplicity reduction alone leaves the original large endpoints in place. The canonical kernel removes $g+1$ vertex-link discs, divides by g , and recovers the fixed core D . It returns exactly g compressing-disc components. For a fixed triangulation and deterministic scheduler, its core orbit trace is independent of the number of bits of g .

This is a concrete improvement for a supplied-vector subroutine. Primitive quadrilateral coordinates may themselves remain large, and the operation does not search for a useful normal vector or construct its knot-exterior provenance. Those separate geometric obligations continue to control the general recognition problem.

5 Certificates, implementation, and the trust boundary

5.1 A layered certificate contract

The implementation separates three assertions that require different evidence. First, a finite pairing system has a particular weighted orbit histogram. Second, that pairing system and those weights actually encode the supplied normal surface. Third, an optional canonical reduction preserves the requested essential-disc count. A correct assertion at one level does not substitute for the assertions above it.

The schema `weighted-interval-orbits-v1` contains the universe size, weight dimension, canonical source weight runs, an unweighted orbit proof, and the claimed histogram. The checker receives the caller’s pairing and weight inputs separately. It reconstructs their canonical forms, checks the unweighted local moves against the actual pairings, and independently transports the weights. A valid unweighted orbit count alone cannot certify the weighted answer: different partitions can have the same number of orbits and different component statistics.

The schema `normal-component-census-v1` adds the original-source fingerprint, query mode, boundary homology basis, and interpreted component summary. Its verifier revalidates the triangulation and coordinates, checks the proposed basis by binary elimination, rebuilds the orbit source, and checks the weighted proof. It then reconstructs the displayed component predicate and multiplicity sums independently of the producer’s result interpreter. Total Euler characteristic provides an additional consistency check, but does not replace componentwise verification.

The outer schema `normal-disc-count-v1` records the vertex-link multiplicities, quadrilateral divisor, explicit core coordinates, core certificate, and final count. Its verifier reconstructs the minimum-triangle subtraction and gcd calculation without calling the producer’s reduction helper. A nonzero divisor requires a disc-mode certificate for the actual reconstructed core. A zero divisor requires a zero residual vector and no core certificate. The final multiplicative count is checked exactly.

Fingerprints make stale evidence and accidental source substitution easy to detect. They are not the entire justification for the result: the replay starts from reconstructed geometry and exact source-dependent pairing and weight data. Neither a matching hash nor agreement on the final component count permits omitting the local checks.

5.2 What independence means here

The producer discovers an orbit-reduction order and implements translation transport by quotient/remainder range additions. The checker validates that order and uses the separate residue-anchor and discrete-derivative method of section 2. It never invokes orbit discovery or the weighted producer’s transport routines. An audit disables the producer entry points before replay to test that separation directly.

There is nevertheless a shared geometric foundation: the existing finite triangulation validator, normal-coordinate validation, integer decoding, and the cell-weight construction. The full-coordinate mode also shares the routine that derives a signature from a recovered component vector. The Regina audit is therefore important: it compares actual expanded connected normal surfaces, constructed by a different implementation, with all three query modes. Agreement is strong finite evidence about those shared routines; it is not a proof assistant verification of the source code. The mathematical proofs establish the algorithmic claims, while executable replay and oracles test the delivered realization of those claims.

The new modules are deliberately additive. They use the maintained `interval_orbits` scheduler and its existing independent local-rule checker, retaining the repository’s classical and Fine–Wilf merger choices. The existing recognizer does not automatically call the new census,

and no legacy result is relabeled as a stronger decision. This makes integration reviewable: a caller must explicitly request the additional operation and interpret its stated source contract.

5.3 Completion, cancellation, and resource limits

A finite orbit allowance is a work limit, not a mathematical negative result. If discovery exhausts `max_cycles`, the new query returns `INCONCLUSIVE`; it does not return the partially accumulated histogram as a complete component census. Cancellation callbacks are passed through input validation, interval construction, discovery and replay, and their exceptions propagate. A caller that supplies an already completed orbit proof cannot simultaneously ask that an orbit-search cycle cap constrain its replay.

Encoded integers remain exact Python integers. The existing integer codec also accepts its explicit hexadecimal representation for certificates and inputs too large for convenient decimal conversion. The large-coordinate experiments retain such values in binary-sized representations; they do not change the meaning of a coordinate or replace it by a floating-point estimate.

Certificate validation is polynomial in the explicitly supplied source and proof lengths. A deployment that reads arbitrary external JSON must impose its own byte and nesting limits before decoding; a replay routine cannot make reading an arbitrarily oversized document free. Those input-container limits are separate from an orbit-cycle cap and from the mathematical bit complexity of a well-formed certificate.

5.4 Using a positive answer in a recognizer

For a source-bound knot exterior, a positive essential-disc count can be used as a positive unknot witness. If a later operation needs the actual surface, request `mode='coordinates'` and select a certified compressing-disc record. Its multiplicity is information about parallel copies, while its coordinate vector describes one connected component. The three-weight result alone need not identify that vector.

For a canonical count query, the simplest extraction strategy is to run the coordinate census on the explicit primitive core. An essential disc in the core is embedded in the same triangulated manifold and is already a sufficient positive witness. Multiplying it back is unnecessary for the existence question. The package does not yet implement a selected-component extraction certificate that avoids the full $7t$ histogram; that is a concrete subsequent optimization.

At the outer boundary, a validated compact manifold with torus boundary is not by itself evidence that it is the complement of a particular knot diagram. A future integration must retain or replay that correspondence. The positive implication is then topological and exact; the absence of a disc in one candidate remains only a candidate rejection. This is the contract under which the new operation can safely become part of a complete recognizer.

6 Reproducible verification and performance evidence

6.1 What was measured and what was held fixed

The experiments use the pinned repository source and the six additive modules in this archive. Recorded hashes identify the actual implementation. The runtime was CPython 3.12.14 on Linux x86-64. Regina’s Python package version was 7.4.1; its engine version string was 7.4. Plot generation used Matplotlib 3.10.8. Timing runs were serial, with no concurrent experiment launched by this work. The shared host was not CPU-pinned, and the measurements are not claims about a dedicated reference machine.

The normal-surface benchmark uses seven paired rounds after one warm-up call per arm, with a fixed random seed and a randomized arm order in every round. Every timed call

validates its source, builds the relevant certificate, and independently verifies it. Fixture construction, process startup, parsing the saved input and final JSON serialization are outside the timed interval. The full-coordinate arm also computes and certifies richer output than the three-weight arm, while both answer the same disc-count question. The reduced-count arm answers that question without claiming a full census of the original vector. These output differences are intentional specializations and should remain visible when comparing the timings.

The tables report median total times. A speed ratio is the median of the seven within-round ratios, not the quotient of the two marginal medians. An identical A/A arm measures ordinary run-to-run variation. No fitted polynomial exponent or asymptotic complexity conclusion is inferred from these finitely many times.

6.2 Independent finite and large-binary oracles

The abstract weighted-kernel audit checks 10,000 seeded finite interval systems against literal union-find orbit sums and against the independent checker. They comprise 299,040 literal points and 282,293 reduction events, including 24,484 translation and 14,172 reflection pushes. Signed overlapping weights, reversals, transmissions, both merger rules, and disconnected supports are exercised. Eight closed-form systems extend the check through 20,000-bit universe sizes without expanding their points. The largest live run list in this finite audit has 18 runs; this observation is evidence about these fixtures, not a substitute for lemma 2.2.

A separately written checker audit uses another deterministic seed and 2,500 literal union-find cases, 366 deliberate certificate mutations, and 81,412 reduction events. It includes a 20,002-bit endpoint case and two producer-disabled replays. All pass. The delivered 103-method regression selection includes the 30 new methods and 73 maintained interval and normal backend methods; all pass from the portable bundle. This is the stated selection, not a claim that the repository’s entire application suite was rerun.

The external audit uses Regina [10] to construct and expand actual normal surfaces. Expected component vectors, Euler characteristics, orientabilities, boundary circles, boundary edge weights and disc predicates come from that external implementation. The saved corpus contains raw face pairings and expected component data, so it can be inspected without relying on a verbal description of the fixtures.

External normal-surface audit	Count
Input normal vectors	1,275
Triangulations, including deterministic relabelings	45
Census comparisons across three query modes	3,825
Census-certificate replays	3,825
Canonical disc-count comparisons	1,275
Canonical disc-count certificate replays	1,275
Failed comparisons or replays	0

The corpus includes 976 disconnected inputs, 252 with nonorientable components, 778 with nontrivial quadrilateral gcd, and 537 with positive vertex-link subtraction. There are 152 triangle-only inputs. It combines layered solid tori, boundary caps with additional vertices, interior vertex links, compatible mixtures of discs, spheres and Möbius bands, finite trefoil and figure-eight exteriors, and relabelings. The largest explicitly expanded surface has 1,004 normal polygons and the largest triangulation has 10 tetrahedra; the configured expansion cap is 4,096 polygons. The large binary tests and the external expanded audit therefore serve complementary purposes.

6.3 Two recorded counterexamples to aggregate tests

The small separate counterexample file isolates cases that can be read and replayed without opening the full corpus. In the one-tetrahedron solid-torus case `1st_1_2:8`, the vector

$$(2, 2, 0, 0; 0, 0, 2)$$

is a union of two essential discs. Their total mod-two boundary class is zero, while the componentwise count is two. The complete audit contains 198 inputs with essential discs but this total-homology cancellation.

In `interior_finite_trefoil:98`, an eight-tetrahedron finite trefoil exterior contains the disjoint normal union of a four-polygon vertex-link sphere and a 33-polygon once-punctured torus. The union is orientable, has total Euler characteristic $2 - 1 = 1$ and nonzero total boundary class, and contains no disc. Six orientable examples in the corpus have this latter combination; ten do so without the orientability restriction. There are 72 inputs with positive total Euler characteristic but no disc component. These cases demonstrate a correctness requirement that faster aggregate arithmetic alone cannot meet.

6.4 Actual normal-surface dimension comparison

For $t \in \{4, 8, 16, 32, 64, 128\}$, the benchmark supplies a primitive meridian disc in a layered solid torus. The maintained fixture has $F_{t+5} - 5$ polygons, where $F_1 = F_2 = 1$. At $t = 128$ this is exactly

$$2\,791\,715\,456\,571\,051\,233\,611\,642\,548$$

normal polygons. Its largest coordinate has only 89 bits. Both arms operate on that compressed source, without building its expanded cell complex.

t	Input bits	Three weights (ms)	$7t$ weights (ms)	Paired ratio	A/A
4	3	2.97	4.39	1.47	1.002
8	6	6.37	13.69	2.17	1.003
16	11	11.66	39.23	3.26	1.006
32	22	58.55	268.36	4.71	1.090
64	44	203.93	1728.48	9.21	0.992
128	89	1152.80	12796.75	11.13	1.004

Table 1: Actual normal-vector disc queries. Total milliseconds include source validation, certificate production and independent replay. The ratio compares full-coordinate extraction with the three-weight query; A/A compares two identical three-weight arms.

The paired advantage increases from 1.47 at four tetrahedra to 11.13 at 128. At the last size, the median times are approximately 12.80 seconds and 1.15 seconds. Both use 133 orbit cycles: the improvement comes from weighted work and result interpretation, not fewer discovery cycles. The observed maximum run lists are 385 for full coordinates and eight for the three statistics. Thus this particular family benefits both from smaller vector dimension and a smaller weight-run representation. The theorem claims the dimension reduction in the relevant arithmetic terms; it does not predict that these wall-clock ratios must equal $7t/3$.

The normal benchmark's A/A paired ratios range from 0.990 to 1.090 across both experiment families. The 32-tetrahedron run exhibits appreciable variation. All samples are retained, and the graph shows their ranges. The large-case improvements comfortably exceed the observed typical A/A variation, but this small experiment is not a distributional study of knot instances or a universal performance guarantee.

6.5 Remove large coordinates even when the original gcd is one

Fix the 16-tetrahedron meridian vector D and boundary vertex link L , and set

$$x_b = 2^b D + (2^b + 1)L, \quad b \in \{64, 512, 4096, 32768\}.$$

The gcd of the original normal coordinates is one. Each input contains exactly 2^b essential discs; the additional vertex-link discs have inessential boundary. The core is always D , whose largest coordinate has 11 bits. The reduced route uses 21 orbit cycles at every size, compared with 39 for the direct three-weight query.

b	Input bits	Direct (ms)	Core (ms)	Paired ratio	A/A
64	75	25.72	13.60	1.87	0.997
512	523	27.80	13.87	2.00	1.011
4,096	4,107	43.80	14.59	3.02	0.999
32,768	32,779	169.70	21.81	7.64	0.990

Table 2: Canonical reduction for $x_b = 2^b D + (2^b + 1)L$. Both routes count the original input’s essential discs and independently verify their evidence. The core always has 11-bit coordinates. A/A compares identical core arms.

At $b = 32768$, the largest input coordinate has 32,779 bits. The median total times are 169.70 ms directly and 21.81 ms through the core, with a paired median ratio of 7.64. Serialized proof sizes are approximately 4,417 KiB and 55 KiB, respectively. The outer core certificate still records large input multiplicities, so it grows with b even though the internal core orbit proof is fixed. Likewise the reduced total runtime is not constant: reading, validating, subtracting, taking the gcd, dividing and binding the original source remain necessary.

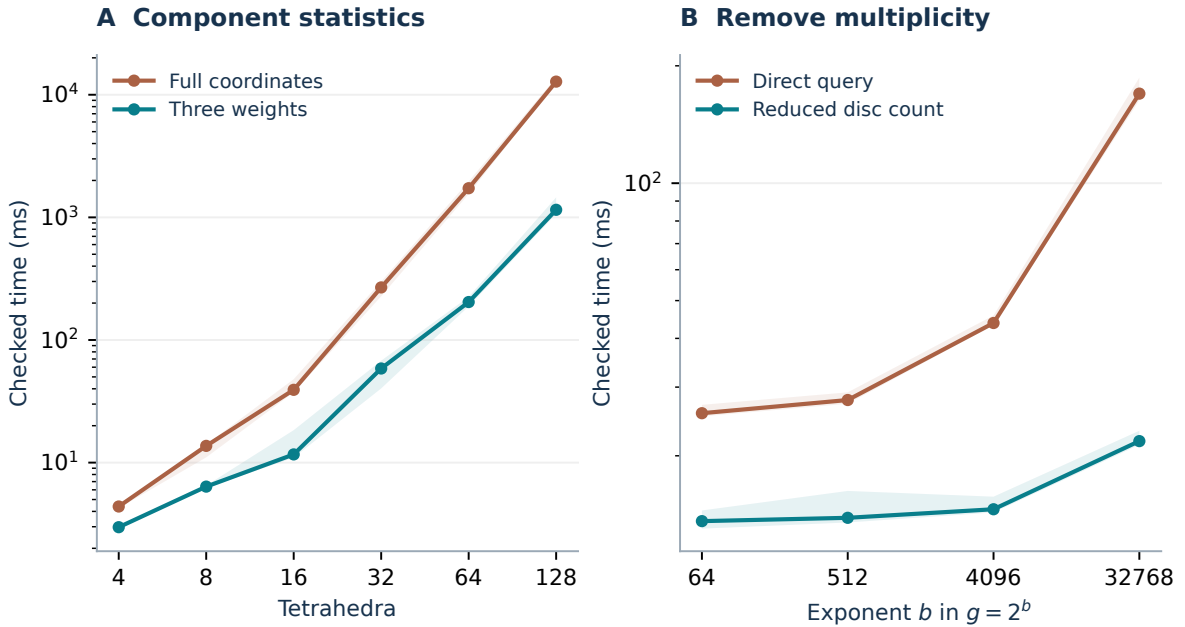


Figure 1: Checked query times from the archived raw samples. Lines show medians; shaded bands span the observed minimum and maximum of the seven rounds and are not confidence intervals. Both horizontal axes and the time axes are logarithmic. Panel A uses supplied normal meridians; panel B uses the fixed-core family of table 2. No curve is an asymptotic fit.

6.6 A controlled abstract dimension experiment

An additional interval-only benchmark holds the endpoint/run structure fixed while changing the transported dimension. It uses translation and reflection families, 32-bit and 4,096-bit scales, and dimensions 28, 112 and 448 against three weights. Its twelve cases have nine arms, nine paired rounds and two warm-ups per arm. The full checked arm projects and merges its answer to the same three-coordinate histogram as the compact checked arm. Its paired ratios range from about 2.0–2.43 at dimension 28 to 18.0–28.34 at dimension 448, with compact A/A ratios 0.935–1.065.

This experiment isolates arithmetic effects more cleanly than the normal family, where coordinate weights also have more runs. Its dimension parameter does not describe a triangulation, so these numbers are not normal-surface or knot-recognition timings. The raw JSON includes smaller-output count-only controls, producer-only times and replay-only times; none of those smaller tasks is reported as a same-task speedup over the weighted query. Only the clean serial run is included in the final evidence.

7 What would establish quasi-polynomial recognition?

7.1 Separate the query bound from the number of queries

Let n be the size of a knot diagram and suppose a source-bound exterior has $t = \text{poly}(n)$ tetrahedra. The results above make one supplied-vector disc query polynomial in t and its coordinate bit length. This remains true if the normal surface contains exponentially many polygons or components. It gives no bound on how many candidate vectors a recognizer must try. An exponential search equipped with a faster polynomial verifier is still an exponential search.

There are at least two useful directions for continuing the project. A search-based route can exploit canonical Q-data, certified pruning, and cheap component predicates. A hierarchy-based route can use the same compressed primitives when surfaces and their complements are transformed. Neither direction may replace its missing progress theorem by favorable timings. The distinction is particularly important for families of layered solid tori: their exponentially large normal discs stress compressed arithmetic, but the benchmark hands the disc vector to the algorithm. It does not measure the difficulty of finding that vector from a knot diagram.

7.2 A precise conditional budget theorem

Lackenby’s Proposition 9.1 bounds hierarchy iterations by $L(g+1)^L$; Proposition 10.2 gives $L \leq 4c_q(\mathcal{H}_1)$ for the refined process [8]. The following elementary bookkeeping formulation makes the desired improvement precise. It is a conditional statement about a future implementation, not a geometric bound established here.

Proposition 7.1 (Sufficient mixed-radix budget). *Suppose a correct recognition algorithm on an input of size n has a lexicographically decreasing integer progress vector $(a_1, \dots, a_L)$ with $0 \leq a_i \leq G_i$, where the G_i are fixed nonnegative integers. Suppose these bounds are valid throughout the entire computation, including suffix reconstruction. Assume that each phase before the first decrease, between successive decreases, and after the last decrease performs at most $\text{poly}(n)$ operations, each costing $\text{poly}(n)$ bit operations on polynomial-size states. If*

$$L = \text{poly}(n), \quad \sum_{i=1}^L \log_2(G_i + 1) = O((\log(n+2))^k),$$

then its running time is $2^{O((\log(n+2))^{\max(1,k)})}$. In particular, $k = 2$ gives $n^{O(\log n)}$ time.

Proof. Use the mixed-radix integer

$$\Phi(a) = \sum_{i=1}^L a_i \prod_{j=i+1}^L (G_j + 1).$$

It is an integer between zero and $\prod_i (G_i + 1) - 1$. Decreasing one digit and replacing any later digits by their largest permitted values still decreases Φ by at least one. Thus there are at most $\prod_i (G_i + 1)$ strict decreases. Multiplying this by the polynomial bound for work between decreases gives

$$\log_2 T(n) \leq O(\log(n + 2)) + \sum_i \log_2 (G_i + 1),$$

which proves the assertion. The same estimate charges polynomial-size certificates and their generation and replay whenever they are included in the per-operation hypothesis. $\square$

Several cautions are mathematically substantive. A linear bound on L and polynomial bounds on every G_i yield only $2^{O(n \log n)}$ by this argument. They do not imply quasi-polynomial time. Many digits with $G_i = 1$ can already cause an exponential product. Nor does a logarithmic hierarchy depth help if the permitted digits have exponential bit lengths. Finally, state-dependent digit bounds need a single valid potential or a proved amortized replacement; substituting small bounds observed on one run is insufficient.

The mixed-radix statement identifies a useful research target. One might bound the total logarithmic digit budget, construct balanced blocks with certified smaller budgets, or prove that the realizable states occupy a much smaller set than the entire product box. The last option requires an actual bound on reachable states and a way to recognize equivalent states within the intended time. A memoization table without such a theorem can itself be exponentially large.

7.3 A compressed transition must preserve its hypotheses

Polynomial verification and polynomial discovery are different obligations. Theorem 14.1 of the 2026 hierarchy paper concerns efficient encoding and checking of an essential hierarchy. Proposition 14.2 supplies a polynomial compressed transition under its uniform-type, irreducibility and essential boundary-pattern hypotheses, with a connected normal surface [8]. The transition caller must establish essentiality for its actual boundary pattern; a knot-exterior triangulation alone does not supply that hypothesis.

The appropriate implementation objective is consequently a checked transition whose caller supplies the relevant established hypotheses, or a broader algorithm with explicit alternative outputs when those hypotheses fail. Likewise, retaining a small description of the parallelity complement is not the same as constructing every sheet of the cut-open triangulation. Our weighted census supplies component data but does not yet retain all attaching maps required by that transition.

Fast compressed curve operations are relevant to the boundary interface. For example, Lackenby's curve algorithms compute geometric intersection and test isotopy in time polynomial in triangulation size and logarithmic curve weights [9]. Their inputs, essentiality assumptions and representation changes must be matched explicitly when used for a hierarchy pattern. The torus homology test in this article is a smaller specialized operation; the higher-genus obstruction in section 3 explains why it cannot replace the more general curve machinery.

8 A prioritized research agenda

The next objective is to turn the proved query $x \mapsto \mathcal{D}(x)$ into a complete recognition procedure with an explicit global complexity bound. The questions below separate near-term

implementation work from structural estimates that remain to be proved. Throughout, n denotes the crossing number of the supplied diagram, not the minimum crossing number of its knot. Each proposed advance should identify its input, its checked output, and the quantity that decreases or pays for its cost.

8.1 First priority: certify the diagram-to-exterior construction

Can the fast diagram pipeline produce a finite triangulation together with efficiently checkable evidence that it represents the exterior of exactly the input knot? A useful certificate should describe the local crossing and arc blocks, their face identifications, and a peripheral marking. Subsequent simplification should carry a sequence of verified local transformations or another explicit homeomorphism certificate. The construction must have a proved polynomial bound on tetrahedra and certificate length in the diagram encoding.

This bridge makes a positive disc query relevant to the original recognition problem. The immediate experiment is to attach provenance to existing diagram-derived fixtures and compare the resulting finite triangulations with an independent topology implementation. Such comparisons guide development; the mathematical obligation is a proof for every accepted construction. Mutation tests should change an arc attachment, a peripheral label, or a simplification step and require the checker to detect the broken correspondence.

8.2 First priority: a complete constrained disc producer

What search procedure supplies the normal vectors, and how many must it inspect? The delivered count eliminates a candidate with $\mathcal{D}(x) = 0$; it gives no exhaustion certificate for the remaining admissible vectors. Develop a producer with a clearly specified finite search space, exact feasibility bounds, and independently justified pruning rules. Record search nodes, candidate bit lengths, and time spent in feasibility, lifting, and component queries separately.

An essential disc can already be found on a Q-vertex ray under the hypotheses of Suh's theorem; proving that existence again is not the research question [4]. The difficult question is whether a complete search can avoid exponentially many relevant branches on every input. The established branch-and-bound framework provides a practical comparison [5]. A proposed pruning rule needs a proof that some essential-disc witness survives whenever one exists, including when an intermediate candidate is disconnected or has nonorientable components.

8.3 First priority: canonicalize before expensive search operations

Can candidate generation operate directly on primitive quadrilateral data, constructing the canonical standard lift only when required? Canonical lifting and vertex-link ambiguity are established Q-theory [2, 3]. The opportunity is to use them earlier in the implemented search, so that large common multiplicities and independent vertex-link offsets never enter its costly orbit queries.

For every proposed Q-coordinate input, construct or certify its integral lift and check the full admissibility conditions. Evaluate whether branching bounds can use the coordinate estimate of Theorem 4.2. The first experiment should compare search-node and arithmetic counts before and after canonicalization on the same candidate stream. The identity $\mathcal{D}(x) = g\mathcal{D}(P)$ supports reuse of an essential-disc count. A request for the original component histogram still requires its original surface data: vertex-link deletion changes that histogram, and scaling a one-sided component changes its sheet decomposition.

8.4 Near-term refinement: certify and benchmark the two-weight mode

The proposition in section 3.6 already proves that one ambient-certified boundary probe suffices for all supplied surfaces in a fixed supported manifold. Implement its finite binary preprocessing and its linear-size cocycle witness, then compare two and three weights on identical candidate streams. The open performance question is the break-even point between additional ambient preprocessing and the saved transport coordinate. Cache only source-verified preprocessing data, and include cold and amortized costs separately. The existing three-weight implementation remains the reference until the new mode passes the frozen external corpus and the source-mutation checks. This is an implementation and cost question, not a conjecture about whether the two-weight criterion is true.

8.5 Second priority: batch related candidates and reuse checked traces

How much computation can be shared across candidates in one triangulation? The canonical core gives an exact starting point: candidates $x_j = \sum_v \ell_{jv} L_v + g_j P$ share the answer $\mathcal{D}(P)$, with their counts recovered by multiplication. Cache the validated triangulation, canonical cores, and their certificates using explicit source identifiers. Charge the cost of reading each candidate and checking its reduction even on cache hits.

A broader question is whether an interval-orbit trace can be reused over a region of coordinate space. Equal quadrilateral support alone does not fix endpoint order or scheduler decisions. A reusable trace would need certified inequalities and congruences defining where every step remains valid, including periodic cases. The completed replay artifacts provide a test harness for this experiment [1]. Measure reuse per distinct core and per verified trace region, and retain the total candidate count in the global cost accounting.

8.6 Second priority: connect selected components to compressed cutting

Can a positive three-weight query return enough additional information to cut along one certified essential disc efficiently? Recover a selected component's normal vector, using coordinate weights or a targeted orbit query, and supply the gluing and boundary data needed by the next stage. This tests a concrete transition from a component predicate to a geometric operation. A histogram with positive disc multiplicity establishes existence, but does not by itself specify the cut triangulation.

Compressed cutting must preserve the hypotheses of the routine being used. Lackenby's hierarchy framework imposes explicit irreducibility and essential boundary-pattern conditions [8]. Determine how each condition is established initially and propagated through subsequent cuts, and include the relevant boundary-pattern evidence in each transition certificate. Efficient algorithms for curves and normalization offer useful subroutines [9]; the surrounding three-dimensional invariants still require proof. Begin with cuts whose outputs have small independent finite oracles before enlarging the binary weights.

8.7 Structural priority: prove an amortized hierarchy budget

Which structural invariant can limit the aggregate search in an accelerated hierarchy? For a proposed encoding with counters ranging from zero to g_i , a useful *target*, not a theorem established here, is

$$\sum_i \log_2(g_i + 1) = O(\log^2 n).$$

These g_i are hierarchy counter bounds, unrelated to the quadrilateral gcd of a single input. If all other state choices are charged within the same budget, the product of the counter ranges is $2^{O(\log^2 n)} = n^{O(\log n)}$.

The proposed bound must control all reachable states relevant to the algorithm, including branching and counter resets. A bound along one chosen successful path is insufficient for an exhaustive search. Seek an invariant that pays for each split, new boundary pattern, and increase in a counter's range. The explicit hierarchy machinery in [8] supplies a precise setting in which to formulate this obligation. Failure to obtain the target should produce an explicit obstruction family or a weaker proved budget, rather than an empirical asymptotic claim.

8.8 Structural priority: anisotropic bounds and a graph of states

Can individual counter bounds replace a uniform worst-case cap? If an encoding has m counters, retain $\prod_{i=1}^m (g_i + 1)$ instead of replacing it immediately by $(1 + \max_i g_i)^m$. This mixed-radix accounting can expose instances with many small choices and only a few large ones. Identify a topological reason for that anisotropy and test whether it persists after cutting and simplification.

Can histories reaching the same certified state be merged into a directed acyclic graph? State equivalence must preserve the manifold, boundary pattern, peripheral information, and remaining search obligations. Prove acyclicity using a well-founded rank, or account explicitly for cycles. Canonical labels and hashes may locate possible matches, but each merge needs a verified equivalence. Initial measurements should count distinct states, transitions, and equivalence-checking cost. A small state graph is useful only when constructing and recognizing its states is also affordable.

8.9 Quantitative priority: an explicit deterministic bit bound

What exponent does the composed algorithm actually satisfy? Derive separate bounds for triangulation construction, candidate production, integer gcds, interval operations, weight-run growth, component recovery, cutting, and certificate checking. Track maximum operand bit lengths and the cost $M(b)$ of multiplying b -bit integers. The weighted AHT theorem gives a polynomial local foundation [1]; composition still requires bounds on the number and size of its invocations.

Publish a deterministic bound with fixed constants, distinguishing $2^{O((\log n)^2)}$ from $2^{O((\log n)^3)}$ and stating which exponent is actually proved. Include output and certificate lengths, memory, and input reading. The completed operation counters and paired kernel measurements can identify costly terms and check predictions. Their role is to guide which inequalities need improvement; the final worst-case bound must follow from the algorithm and its invariants.

8.10 Continuous priority: stronger adversarial evidence

Extend the delivered 1,275-vector, 45-triangulation Regina corpus and its isolated counterexamples. Preserve both essential-disc cancellation and the orientable sphere-plus-punctured-torus false positive as mandatory cases. Add large vertex valence, additional boundary vertices, long layered triangulations, coprime large quadrilateral coordinates, and inputs designed to increase interval subdivisions. Exercise one-sided scaling and certificate mutations at every new interface.

Keep a capped corpus with fully expanded independent components and a separate collection of large binary instances whose answers follow from proved constructions. For each family, report the parameter varied, the expected component semantics, the source triangulation, and reproducible seeds. When recognition stages are connected, add difficult diagram families and measure every stage on the same instances. This prevents a local speedup from hiding a newly dominant producer or source-validation cost, while maintaining the exact distinction between a component of a supplied surface and a conclusion about the original knot.

9 Reproduction and repository integration

9.1 Archive organization

The archive contains this PDF, the complete $\text{\LaTeX}$ source and generated figures, a portable Python source snapshot, the additive integration patch, raw audit and benchmark JSON, the frozen external corpus, two explicit counterexamples, and a manifest of source identities and file hashes. The top-level `README.md` gives the short entry points; `INTEGRATION.md` describes the API contracts and the intended review. The baseline commit is recorded in full so the work can be compared with later changes to the repository.

The runnable snapshot is under `code/fast`. It preserves the entire `fastunknot` package because the maintained package initializer imports the existing recognizer infrastructure. Copying only a few local modules would not preserve ordinary package imports. The snapshot includes only the selected regression files and the fixture dependencies they require; it does not carry unrelated historical benchmark directories. The original MIT-0 license and the internal attribution files are retained.

The new functionality consists of six modules: the weighted producer and independent verifier, the geometric weight construction, the component census and verifier, and the canonical disc-count wrapper. Two new test files and four audit or timing drivers accompany them. These additions do not modify the maintained recognizer’s defaults. The integration patch is intended to be applied at the repository root after checking the pinned dependency identities; the full snapshot is for reproduction, not an instruction to overwrite a newer checkout.

9.2 Commands and optional dependencies

Python 3.10 or newer is required by the maintained package. The weighted and native finite-geometry code uses the standard library. Regina is needed for the external normal-surface audit and the Regina-specific test methods; those unit methods are reported as skipped if it is absent. Matplotlib is needed only to regenerate the figures. A $\text{\LaTeX}$ installation with the packages named in the main source is needed only to rebuild the PDF.

From the unpacked archive, use:

```
python3 reproduce.py tests
python3 reproduce.py audit
python3 reproduce.py regina
python3 reproduce.py benchmark
python3 reproduce.py figures
python3 reproduce.py paper
python3 reproduce.py verify-manifest
```

The benchmark command runs the recorded full experiment sizes, so it takes longer than the finite audits. The launcher uses the current Python interpreter and supplies the import paths explicitly. Recomputed results go to `reproduced/`, preserving the archived measurements in `results/`. The Regina command reads the frozen corpus instead of reenumerating vertex solutions, reducing sensitivity to enumeration order in another Regina release. The generator can also construct a new corpus when that is the intended experiment.

Tables and plots are generated from raw results by `article/make_figures.py`; the corresponding CSV is included. A paper rebuild uses the archived figures and tables unless they are explicitly regenerated. The main $\text{\LaTeX}$ file uses a self-contained bibliography, so BibTeX is unnecessary. The manifest check compares bytes; it does not claim that independently rerun timing JSON or PDF timestamps must match byte for byte.

9.3 Minimal native use

The following example runs entirely on the included finite fixture. Its expected count is two even though the total boundary class is zero.

```
from normal_orbit_research.fixtures import layered_torus
from fastunknot.normal_surface_components import normal_component_census
from fastunknot.normal_component_verify import (
    verify_normal_component_certificate,
)

tri, meridian = layered_torus(4)
vector = [[2 * value for value in row] for row in meridian]
answer = normal_component_census(tri, vector,
                                record_certificate=True)
assert answer['compressing_disk_components'] == 2
assert verify_normal_component_certificate(
    tri, vector, answer['certificate'])
```

The separate executable example also creates a large input with coordinate gcd one, verifies its canonical reduction, and saves both proof objects. This gives an integration reviewer an immediately inspectable positive case, an independent replay path, and explicit evidence of the source contract.

9.4 Acceptance criteria for the next integration

A reasonable first merge criterion is preservation of the old normal and interval APIs, successful replay of the saved corpus, rejection of the recorded source and histogram forgeries, and a caller-facing distinction between a complete query and an inconclusive work limit. A later recognizer integration should additionally specify the provenance witness and the candidate-generation contract before using a positive disc count as a knot decision. The benchmark scripts should remain usable as those later stages are introduced, with separate measurements for discovery, transport, reconstruction and verification.

The concrete result available now is a compressed, checked operation that was missing from the native backend, together with a smaller exact input for its essential-disc count. The next mathematical milestone is a bound on the complete search or hierarchy that calls such operations, with every representation change and bit cost charged.

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